



PROJECT REPORT

SUPERMARKET SALES ANALYSIS

Interactive Data Analysis and Power BI Dashboard

Submitted to

ANUDIP FOUNDATION

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1. Executive Summary

The Supermarket Sales Analysis project is a Business Intelligence project developed using Microsoft Power BI. The objective is to convert transaction-level sales information into an interactive dashboard that makes sales, profitability, product performance, geographic performance, customer segments and payment methods easier to understand.

The dashboard combines Key Performance Indicators (KPIs), charts, geographic analysis and interactive slicers. Users can filter the report by region and observe how the displayed measures and visualizations change. This makes the report suitable for quick performance monitoring as well as exploratory analysis.

The final deliverables for the project include the Power BI dashboard, presentation material and this project report, which together document the purpose, data fields, methodology, dashboard design and observations.

Project at a Glance

Item	Details
Project Title	Supermarket Sales Analysis
Organization	Anudip Foundation
Team Members	Tuna Santra; Mainak Das
Primary Tool	Microsoft Power BI
Project Type	Business Intelligence / Data Visualization
Main Output	Interactive Sales Analysis Dashboard
Key Interaction	Region slicer

2. Introduction

Retail and supermarket operations generate large volumes of transactional information. A raw dataset can contain useful information about orders, products, locations, customers and payments, but the information is difficult to interpret when viewed only as rows and columns.

Data visualization provides a way to transform such information into meaningful summaries. In this project, Microsoft Power BI is used to create a single dashboard where important measures can be monitored and compared through interactive visuals.

2.1 Problem Statement

The main problem addressed by the project is the difficulty of obtaining a clear and interactive view of supermarket sales performance from raw transaction data. Management or analysts need to identify sales trends, profitable areas, strong-performing products, geographic differences and customer/payment patterns without manually inspecting individual records.

2.2 Proposed Solution

The proposed solution is an interactive Power BI dashboard that summarizes the dataset through KPI cards, category and sub-category charts, geographic visuals, customer segment analysis, payment-method analysis and time-based trends. A Region slicer provides an interactive way to filter the report.

2.3 Scope

- Analyze sales, profit and quantity measures.
- Analyze product categories and sub-categories.
- Analyze state and region-level sales performance.
- Analyze customer segments.
- Analyze payment methods.
- Analyze monthly sales and profit trends.
- Provide interactive filtering using a Region slicer.
- Present the results through a professional Power BI dashboard.

3. Objectives of the Project

The project was developed with the following objectives:

1. Create a clear and easy-to-understand view of supermarket sales performance.
2. Summarize key measures such as Sales, Profit, Quantity and Delivery-related information.
3. Compare sales performance across product categories and sub-categories.
4. Identify differences in performance across states and regions.
5. Understand the contribution of different customer segments.
6. Analyze the distribution of sales across payment methods.
7. Track monthly sales and profit trends.
8. Provide interactive filtering so users can explore the dashboard according to region.
9. Create a dashboard that can be presented to a non-technical audience.

3.1 Expected Benefits

- Faster access to important sales indicators.
- Improved visibility into product and geographic performance.
- Easier identification of trends and areas requiring further investigation.
- Interactive exploration instead of relying only on static tables.
- A consolidated view of multiple business dimensions in one report.

4. Dataset Description

The supplied Power BI report uses a sales dataset containing fields for transactions, dates, financial measures, products, geography, customer segments, payment methods and delivery information. The report definition references the main data entity as Table2.

Field	Purpose in Analysis
OrderID	Identifies an order/transaction record.
Order Date	Supports monthly and yearly time analysis.
Sales	Measures sales/revenue generated.
Profit	Measures profit generated.
Quantity	Measures units/quantity sold.
Category	Groups products into major categories.
Sub-Category	Provides more detailed product-level grouping.
State	Supports geographic/state-level analysis.
Region	Supports regional analysis and the dashboard slicer.
Segment	Represents customer segment.
Payment Method	Shows the method associated with transactions.
AvgDelivery	Represents delivery-related performance information.

4.1 Data Dimensions Used

- Time dimension: Order Date
- Product dimensions: Category and Sub-Category
- Geographic dimensions: State and Region
- Customer dimension: Segment
- Payment dimension: Payment Method
- Measures: Sales, Profit, Quantity and AvgDelivery

4.2 Data Preparation

The dashboard requires fields to be interpreted correctly in Power BI. Date fields are used for time-based analysis, numeric fields are used for aggregation, and categorical fields are used for grouping and filtering. Before final presentation, the data should be checked for missing values, incorrect data types, duplicate records where relevant, and inconsistent category or region labels.

5. Technology and Methodology

5.1 Technology Used

Technology / Component	Purpose
Microsoft Power BI	Data visualization, dashboard creation and interactive analysis.
Power BI Data Model	Organizes the fields used by the report.
Power BI Visuals	Displays KPIs, trends, category, geographic and distribution analysis.
Slicers	Allows users to interactively filter the report.
GitHub	Project versioning and submission of project files/documentation.
Microsoft Word	Preparation of project documentation/report.
Microsoft PowerPoint	Project presentation.

5.2 Project Methodology

10. Understand the project requirements and identify the business questions.
11. Inspect the available sales fields and determine how they should be analyzed.
12. Prepare the data so that dates, measures and categorical fields can be used correctly.
13. Load the data into Power BI and organize the required fields.
14. Create summary measures and aggregations for sales, profit, quantity and delivery.
15. Build visualizations for product, geographic, customer, payment and time analysis.
16. Add the Region slicer to provide interactive filtering.
17. Arrange and format the dashboard for readability.
18. Validate that visuals respond appropriately to filtering.
19. Prepare the presentation and project report and organize the deliverables for GitHub.

6. Dashboard Design and Components

The dashboard is designed as a single-page analytical view. A dark visual theme is used, with KPI cards near the top, distribution visuals on the left, trend analysis in the center, geographic analysis on the right, and product/category analysis in the lower and right sections.

6.1 Original Dashboard



Figure 1: Original Supermarket Sales Dashboard developed for the project

6.2 Dashboard Components

Component	Purpose
Sales KPI	Displays the summarized sales value.
Quantity KPI	Displays the total quantity represented by the selected data.
Profit KPI	Displays the summarized profit value.
Delivery Date KPI	Displays the delivery-related value used in the dashboard.
Sales by Segment	Shows the distribution of sales across customer segments.
Sum of Payment Mode	Shows the distribution of sales by payment method.
Monthly Sales by YoY	Compares monthly sales across years.
Monthly Profit by YoY	Compares monthly profit across years.
Sales by State	Shows state-level sales performance.
State Map Area	Provides a geographic visual area; map rendering depends on Power BI map settings.
Sales by Category	Compares sales among major product categories.
Sales by SubCategory	Compares sales among detailed product sub-categories.
Region Slicer	Filters the report by Central, East, South or West.

7. Interactivity and Slicers

Interactivity is an important part of the dashboard. Instead of presenting a fixed set of numbers, the report allows the user to change the analysis context using the Region slicer.

7.1 Region Slicer

The dashboard contains four region options: Central, East, South and West. Selecting a region changes the context of the dashboard visuals so that the user can examine the corresponding sales performance.

7.2 Why Slicers Are Useful

- They reduce the need to create separate dashboards for every region.
- They allow users to compare different business contexts interactively.
- They make the dashboard more useful for exploratory analysis.
- They allow the same visual layout to answer multiple questions.

7.3 User Interaction Flow

20. Open the Power BI dashboard.
21. Select a region from the slicer.
22. Observe the KPI cards and charts update according to the selected filter context.
23. Compare product, state, segment, payment and time-based visuals.
24. Change the region selection to explore another part of the dataset.

8. Dashboard Observations

The following observations are based on the dashboard screenshot supplied for this report. The screenshot shows the West region selected, so the KPI values shown below describe that displayed filter state rather than necessarily representing the entire underlying dataset.

Dashboard Area	Displayed Observation
Sales KPI	Approximately 24.02K in the displayed dashboard state.
Quantity KPI	176 in the displayed dashboard state.
Profit KPI	Approximately 3K in the displayed dashboard state.
Delivery KPI	Displayed value is 3.
Customer Segment	Consumer is the largest displayed segment at 44%, followed by Home Office at 37% and Corporate at 19%.
Top visible states	Washington is approximately 8.0K, Colorado approximately 4.8K and California approximately 3.8K.
Category	Furniture is approximately 12K, Technology approximately 10K and Office Supplies approximately 2K.
Sub-category	Tables is approximately 7.1K, Copiers approximately 6.3K and Phones approximately 3.2K.

8.1 Trend Analysis

The monthly sales and monthly profit visuals compare values by month across the displayed years. These charts make it possible to identify months with relatively higher or lower performance and to compare year-over-year patterns.

8.2 Important Interpretation Note

Dashboard values should always be interpreted together with the active filter selection. A value visible on one screenshot should not automatically be described as the overall value for the complete dataset unless the report is explicitly in an unfiltered state.

9. Results and Discussion

The project demonstrates how a collection of sales records can be transformed into a compact business intelligence dashboard. The dashboard brings together several perspectives that would otherwise require multiple separate tables or charts.

9.1 Product Performance

The category and sub-category visuals provide two levels of product analysis. Category-level analysis gives a high-level comparison, while sub-category analysis helps identify specific product groups contributing to sales.

9.2 Geographic Performance

State and region analysis provides a geographic perspective. This allows the user to investigate whether sales are concentrated in particular states or regions and to compare performance after applying the Region slicer.

9.3 Customer and Payment Analysis

The segment and payment-method visuals add customer-behavior context to the financial measures. Together, they help answer questions about which customer groups and payment methods are represented in the selected data.

9.4 Time-Based Performance

The monthly sales and profit visuals provide a time-series view. Year-over-year comparison can reveal recurring patterns, changes in performance and periods that may require further investigation.

10. Limitations and Future Improvements

10.1 Current Limitations

- The analysis is dependent on the quality and completeness of the source dataset.
- Some dashboard labels are compact because the report is designed to fit many visuals on one page.
- Map and filled-map rendering may depend on Power BI map settings and environment configuration.
- The current dashboard focuses primarily on descriptive analysis rather than predictive forecasting.
- The dashboard screenshot represents one filter state and should not be treated as the complete dataset summary.

10.2 Future Improvements

- Add a dedicated overview page and separate detailed analysis pages.
- Add more slicers such as Category, Segment, Payment Method and Date.
- Add profit-margin KPIs alongside absolute profit.
- Add drill-through pages for state and product analysis.
- Add tooltips containing additional measures.
- Add year, quarter and month drill-down functionality.
- Add forecasting or trend projection if sufficient historical data is available.
- Add data-quality monitoring for missing or inconsistent records.

11. Conclusion

The Supermarket Sales Analysis project demonstrates the practical use of Business Intelligence and data visualization for transforming sales data into an interactive analytical dashboard. Microsoft Power BI was used to organize the available sales fields and present them through KPIs, product analysis, geographic analysis, customer segmentation, payment analysis and time-series visuals.

The Region slicer adds an interactive layer that allows users to examine different geographic contexts without changing the dashboard structure. The resulting dashboard provides a concise way to monitor performance and identify areas for deeper analysis.

The project also provides a foundation for future enhancements such as additional filters, drill-through analysis, profit-margin metrics and predictive features.

12. Suggested GitHub Repository Structure

For submission and version control, the project can be organized in GitHub using the following structure:

```
Anudip-Supermarket-Sales-Analysis/  
├── PowerBI/  
│   └── Supermarket_Sales.pbip  
├── Documentation/  
│   └── Supermarket_Sales_Project_Report.docx  
├── Presentation/  
│   └── Supermarket_Presentation_Final2.pptx  
├── Screenshots/  
│   └── supermarket_sales_dashboard.png  
└── README.md
```

12.1 README Contents

- Project title and short description.
- Project objectives.
- Tools and technologies used.
- Dataset description.
- Dashboard screenshot.
- Main dashboard features.
- Team members: Tuna Santra and Mainak Das.
- Instructions for opening the Power BI project.
- Link/reference to the project report and presentation